

Python Basics — Chapter 1

Introduction to Python Programming

Learning Objectives

By the end of this chapter, you will be able to:

- Explain what Python is and where it is used.
- Write and run a simple Python program.
- Use `print()` to display information.
- Create variables and understand basic data types.
- Work with strings and numbers.
- Use basic arithmetic operators and comments.

1. What Is Python?

Python is a high-level, general-purpose programming language. It is designed to be readable and relatively easy to learn. Python is used for web development, automation, data analysis, artificial intelligence, machine learning, scientific computing, and many other tasks.

Example:

```
print("Hello, Python!")
```

2. Your First Python Program

A Python program is a set of instructions that Python executes. One of the simplest instructions is `print()`, which displays a value on the screen.

Example:

```
print("Hello World")  
print("I am learning Python.")
```

When this program runs, the output is:

```
Hello World  
I am learning Python.
```

3. The `print()` Function

The `print()` function is commonly used to display text, numbers, and the values stored in variables.

Examples:

```
print("Pakistan")  
print(25)  
print(10 + 5)
```

The last statement prints the result of the calculation, so the output is 15.

4. Variables

A variable is a name used to store a value. In Python, you can create a variable by assigning a value with the = operator.

Examples:

```
name = "Anmol"  
age = 20  
score = 95
```

The variable name stores text, while age and score store numbers. You can use variables later in your program.

```
name = "Anmol"  
print(name)
```

5. Basic Data Types

A data type describes the kind of value stored in a variable. Some common Python data types for beginners are:

- **str** — text, such as "Hello"
- **int** — whole numbers, such as 10 or -5
- **float** — decimal numbers, such as 3.14
- **bool** — True or False

Examples:

```
student_name = "Ali"  
age = 18  
height = 5.8  
is_student = True
```

6. Strings

A string is a sequence of characters used to represent text. Strings are usually written inside single quotes or double quotes.

Examples:

```
city = "Karachi"  
message = 'Welcome to Python'
```

You can join strings using the + operator:

```
first_name = "Ali"  
last_name = "Khan"  
full_name = first_name + " " + last_name  
print(full_name)
```

7. Numbers and Arithmetic Operators

Python can perform common mathematical operations. The main arithmetic operators are:

- `+` Addition
- `-` Subtraction
- `*` Multiplication
- `/` Division
- `//` Floor division
- `%` Modulus (remainder)
- `**` Exponentiation

Examples:

```
a = 10
b = 3
print(a + b)
print(a - b)
print(a * b)
print(a / b)
print(a % b)
```

8. Comments

Comments are notes written in code for humans. Python ignores comments when it runs the program. A single-line comment begins with `#`.

```
# This is a comment
name = "Sara" # Store the name
```

9. Checking a Data Type

Python provides the `type()` function to check the type of a value or variable.

```
age = 20
print(type(age))

name = "Ali"
print(type(name))
```

10. Practice Exercises

Try these exercises without looking at the answers first.

- 1 Print your name on the screen.
- 2 Create a variable called age and store your age in it.
- 3 Create two number variables and print their sum.
- 4 Create a variable containing your favorite hobby and print it.
- 5 Calculate the area of a rectangle using length and width variables.

11. Mini Project

Create a small program that stores a student's name, age, and marks, then displays all three values.

Example solution:

```
student_name = "Anmol"
age = 20
marks = 85

print("Name:", student_name)
print("Age:", age)
print("Marks:", marks)
```

12. Chapter Summary

In this chapter, you learned that Python is a general-purpose programming language. You practiced `print()`, variables, strings, numbers, basic data types, arithmetic operators, comments, and `type()`. These concepts form the foundation for writing larger Python programs.

Quick Review

- Python programs contain instructions that Python executes.
- `print()` displays information.
- Variables store values.
- Strings store text.
- `int` and `float` represent numbers.
- `bool` represents `True` or `False`.
- `#` starts a single-line comment.
- `type()` can be used to inspect a value's data type.